

Beyond Naphtha: Capital Allocation, Feedstock-Switching Economics & Forex-Risk Quantification for Reliance O2C's Ethane-Cracking Pivot

SITUATION

- **Reliance's O2C engine is being squeezed by a margin problem, not a volume one.** The O2C segment earned **₹6,26,921 crore revenue in FY25 (+11% YoY)**, yet EBITDA fell to **₹54,988 crore** as weak transport-fuel cracks and muted downstream chemical deltas compressed polymer and polyester margins — driven by record global petrochemical oversupply, with excess capacity in the six key building blocks reaching ~218 million tonnes in 2023, nearly 3× the long-run average.^{[1][6]}
- **The structural fix is feedstock, not price.** Cracking **ethane yields ~80% ethylene per molecule versus ~30% for naphtha**, and ethane is roughly **half as expensive as naphtha on an energy-equivalent basis**. Q3 FY26 already proved the upside: O2C EBITDA rose **15% YoY to ₹16,507 crore**, with favourable ethane-cracking economics explicitly cited as a profit driver.^{[2][3]}
- **The pivot is being built right now.** Reliance has committed **>\$2 billion** to cryogenic ethane terminals at Jamnagar and Dahej (each >1.5 MMTA), a planned ~100 km Dahej pipeline, and **three new very-large ethane carriers** on top of its existing fleet of six. QatarEnergy's shift to supplying "lean" gas makes imported US ethane a deliberate, paid-for feedstock choice rather than a by-product.^{[3][4]}
- **Every substituted tonne is simultaneously a margin and a forex decision.** Each tonne of ethane replacing naphtha shifts exposure from imported crude (India's oil-import dependence **~88.6% in FY25**) to US Mont Belvieu ethane plus freight and USD/INR. US ethane net exports are set to rise **14% in 2025 and 16% in 2026**, even as a 2026 naphtha spike toward \$1,000/t widens the switching prize. Which crackers convert first, at what capex, and hedged how, is a capital-allocation and risk problem.^{[5][7][8]}

THE THREE-LEVER FRAMEWORK

Lever 1: Cracker-Conversion Investment Case (Which Units, Which Feedstock)

Ethane-handling capacity is a per-asset capex decision under uncertainty: which of Reliance's cracker complexes — Jamnagar, Dahej, Hazira, Nagothane, Vadodara — justify incremental ethane-conversion and terminal/pipeline capex first, ranked by current naphtha exposure, ethylene-yield uplift, throughput and proximity to import infrastructure.^[3] Output: NPV, IRR and payback per cracker tier, plus a phased FY27–FY29 capex roadmap sized against the >\$2 billion terminal commitment.^[4]

Our role: Build the Feedstock Stratification Matrix — the ROI-ranked conversion tiers with NPV, IRR and payback per asset.

Lever 2: Cross-Feedstock Margin & Delta Simulation (The Margin Engine)

A flexible cracker only pays if the switching rule does. We estimate the **naphtha–ethane switch-point** — the spread (delta) at which moving a molecule to ethane adds contribution margin net of freight and conversion cost — and simulate the product slate (PE, PP, polyester chain) against China import-parity pricing under persistent oversupply.

Our role: Deliver the Dynamic Delta Simulator — feedstock-mix and product-slate margin recommendations by plant, with explicit switch-point and volume–margin trade-off curves.

Lever 3: Feedstock-and-Forex Risk Quantification (The Hedge)

Each shift trades Brent/naphtha exposure for US ethane, VLEC freight and USD/INR. **Monte Carlo simulation** across ethane price, the naphtha–ethane spread, shipping rates, throughput variance, forex and policy/tariff timing yields a **P10/P50/P90 earnings range** per rollout scenario; the national stakes — ~88.6% crude-import dependence^[5] — scale down to a per-cracker import-substitution value.

Our role: Run the Monte Carlo Risk Simulation to quantify the risk-adjusted value of converting early versus waiting for ethane economics and tariffs to settle, and consolidate all three levers into the Executive Dashboard for O2C leadership.

DELIVERABLES (20 HRS/WEEK, 8 WEEKS, 4-MEMBER TEAM)

- **Feedstock Stratification Matrix.** Ranks Reliance's cracker complexes into conversion tiers by naphtha exposure, ethylene-yield uplift, throughput and infrastructure proximity.
- **Dynamic Delta Simulator.** Interactive model linking the naphtha–ethane spread, ethylene-yield economics and product-slate pricing to plant-level contribution margin.
- **Monte Carlo Risk Simulation.** 10,000-scenario simulation of earnings volatility across ethane/naphtha prices, crude/forex movements and policy-pace assumptions, with downside (VaR-style) metrics.

- **Executive Dashboard.** CFO-level view consolidating the conversion roadmap, NPV/IRR/payback by tier, recommended feedstock mix and risk-adjusted investment recommendations.

RESULT

Reliance O2C gets four things it does not have today: an **ROI-ranked map of which crackers to convert to ethane first**, a **margin tool** that turns the naphtha-to-ethane choice into a live decision rather than a procurement reflex, a **quantified feedstock-and-forex risk envelope** around the pivot, and a **first-mover capital-allocation playbook** for the ethane economy the company is already building toward.^[3]

References:

- [1] [Reliance Industries Limited, Financial Results Presentation FY2024-25 \(25 April 2025\) — O2C revenue ₹6,26,921 cr, EBITDA ₹54,988 cr.](#)
- [2] [Reliance Industries Limited, Media Release: Q3 FY2025-26 Financial & Operational Performance, 16 January 2026 — O2C EBITDA ₹16,507 cr \(+15% YoY\), ethane-cracking economics.](#)
- [3] [Business Standard \(Bloomberg Opinion\), “Mukesh Ambani bets big on US ethane amid shifting global trade dynamics”, 7 July 2025 — ethane vs naphtha yield \(~80% vs ~30%\) and cost, 6 VLECs + 3 planned, ~100 km Dahej pipeline, QatarEnergy lean-gas shift.](#)
- [4] [U.S. Energy Information Administration / Reuters coverage of US ethane exports & infrastructure, 2025 — >\\$2 bn Jamnagar & Dahej cryogenic ethane terminals \(>1.5 MMTPA each\).](#)
- [5] [OilPrice.com / PPAC, “India’s Oil Import Dependence Climbs to Nearly 89%”, 2025 — 88.6% crude-import dependence, FY25.](#)
- [6] [S&P Global Commodity Insights, “Oversupply weighs on global petrochemicals”, 2024 — ~218 Mt excess capacity in six key building blocks \(2023\).](#)
- [7] [U.S. Energy Information Administration, “U.S. ethane exports are expected to grow through 2026” — net exports +14% \(2025\), +16% \(2026\).](#)
- [8] [Market Minute / FinancialContent, “Naphtha Surges to \\$1,000: The Petrochemical Crisis of 2026 Explained”, 30 March 2026.](#)

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